

Data Analytics: Foundations, Modeling, and Intelligent Decision Systems

Course Description This course provides a comprehensive introduction to data analytics, guiding students from foundational concepts through statistical modeling, hypothesis testing, predictive and prescriptive analytics, and the integration of machine learning and artificial intelligence. Emphasis is placed on analytical reasoning, evidence-based decision making, ethical responsibility, and real-world application.

Week 1 — Foundations of Data Analytics Topics: Data vs information, analytics concepts, lifecycle overview.

Week 2 — Analytics Types & Decision Context Topics: Descriptive, diagnostic, predictive, prescriptive analytics.

Week 3 — Data Collection & Sources Topics: Primary vs secondary data, experimental vs observational data.

Week 4 — Data Quality & Cleaning Topics: Data quality dimensions, missing data, standardization.

Week 5 — Exploratory Data Analysis: Univariate Topics: Distributions, central tendency, variability.

Week 6 — Exploratory Data Analysis: Relationships Topics: Bivariate, multivariate, temporal analysis.

Week 7 — Statistical Modeling Topics: Regression, assumptions, bias–variance tradeoff.

Week 8 — Hypothesis Testing & Experimental Inference Topics: Hypothesis testing, McNemar's test, A/B testing.

Week 9 — Predictive Analytics Topics: Forecasting, validation, uncertainty.

Week 10 — Prescriptive Analytics Topics: Optimization, decision trade-offs.

Week 11 — Decision Communication & Deployment Topics: Decision memos, monitoring, drift.

Week 12 — Analytics, Machine Learning & AI Topics: ML models, AI systems, ethics.